Data Management and preparation

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Objectives:

- Identify internal and external data sources.
- Distinguish structured vs. unstructured data.

Data management and preparation

- the process of collecting, cleaning, organizing, and securing data to make it ready for accurate business analysis and decision-making.
- Data preparation is the critical foundation of business analytics. Raw organizational data must be processed systematically before it can yield reliable business insights.

The ETL (Extract, Transform, Load) framework

- To prepare data for analysis, organizations rely on ETL techniques to clean anomalies and standardize datasets.

The ETL (Extract, Transform, Load)

- Extraction: Pulling raw records from multiple internal and external systems.
- Transformation: The core cleaning phase.

This includes:

* Fixing formatting differences across databases.
* Handling missing information or blank values.
* Removing duplicate records to prevent skewed analysis.

- Loading: Moving the newly cleaned data into its final destination.

Data Sources – Internal vs. External

- Analytic systems must ingest data from both inside and outside enterprise boundaries to optimize performance tracking and build strategic frameworks like the Balanced Scorecard [Etgar & Wibisono, 2023].

Internal Data Sources

- Operational data generated directly from within the organization's own day-to-day business processes [Van Looy & Shafagatova, 2016].

Examples:

- Enterprise Resource Planning (ERP) systems, financial ledgers, transactional logs, and HR employee databases.

External Data Sources

- Data acquired from the external market environment surrounding the business.

Examples:

- Industry benchmarks, competitor market share reports, macroeconomic trends (inflation, interest rates), and public social media metrics.

Structured vs. Unstructured Data

- Modern analytics requires dealing with data of vastly different shapes, formats, and complexities.

Structured Data

- is highly organized information that conforms to a rigid, predefined organizational schema or tabular model, making it predictable, easily searchable, and natively compatible with relational databases [Simon, 2014]. Because of its neat and uniform architecture, it can feed directly into traditional business intelligence reporting systems with minimal processing required [Simon, 2014].

Examples of Structured Data

- **SQL databases** containing customer profiles
- **Financial spreadsheets** tracking monthly revenues
- **Barcode inventory** logs in warehouses
- **Flight reservation details** like times and seat numbers
- **Online form submissions** for phone numbers and zip codes

Unstructured Data

- refers to qualitative information that lacks a distinct, predefined structural framework or data model. Typically comprising the largest portion of modern corporate data pools, this data type is inherently heavy and messy, making it incompatible with traditional analytical tools. Consequently, extracting value from unstructured data requires specialized semantic approaches and advanced machine learning models to capture metadata, uncover underlying context, and transform raw content into measurable Key Performance Indicators (KPIs) [Maté et al., 2017].

Examples of Unstructured Data

- **Customer emails** containing qualitative feedback and support requests
- **Phone audio recordings** from customer service centers
- **Corporate PDF reports** featuring free-form text and mixed media
- **Social media videos** and image-based marketing content

Thank you!